LESSON 02 · CHEAT SHEET

Decisions & Loops

IF / ELIF / ELSE

```
if quests_done < 3:       # colon!
    rate = 0              # 4 spaces in
elif difficulty >= 4:     # only if above was False
    rate = 0.25
elif difficulty == 3:
    rate = 0.10
else:                     # none of the above
    rate = 0
print(rate)               # back out: after the if
```

First true branch runs, the rest are skipped. `elif` and `else` are optional.

CONDITIONS

`==` `!=`	equal, not equal (VBA: `=` `<>`)
`<` `<=` `>` `>=`	as in Excel
`a <= x < b`	between (chained)
`and` `or` `not`	combine
`text == "gone"`	compare text, case matters

FOR: ONE ITEM AT A TIME

```
for hero in "Wren,Bram".split(","):
    print(hero)           # Wren, then Bram

for month in range(1, 13):
    print(month)          # 1 … 12  (13 not included)

for record in records.split(";"):
    name, reward, diff = record.split("/")
```

`range(5)`	0 1 2 3 4
`range(1, 6)`	1 2 3 4 5
`range(0, 20, 5)`	0 5 10 15

VBA `For Each x In …`. No counter, no `Next`.

WHILE: UNTIL SOMETHING CHANGES

```
months = 0
while bounty < target:
    bounty *= 1.03        # MUST move towards the end
    months += 1
print(months)
```

Runs for ever? **Ctrl+C** in the terminal. VBA: `Do While … Loop`.

THE LOOP RECIPE

```
total = 0                 # 1. set up BEFORE
count = 0
highest = 0
for r in rewards:
    total += r            # 2. update INSIDE
    count += 1
    if r > highest:
        highest = r
print(total / count)      # 3. use AFTER
```

`x += 1`	`x = x + 1`
`x -= 1`, `x *= 2`	same idea

STEERING A LOOP

```
for x in items:
    if x == "gone":
        continue          # skip to next item
    if x == "STOP":
        break             # leave the loop now
    ...
```

USEFUL IN CONDITIONS

```
day % 15 == 0             # every 15th: True on 15, 30
text.lower() == "yes"     # case-insensitive compare
if years == 1:   unit = "year"
else:            unit = "years"
```

GOTCHAS

- Missing **colon** after `if` / `for` / `while` / `else`
- `if x = 3:` → `==`
- Mixed tabs/spaces: use 4 spaces, always
- `total = 0` inside the loop resets it every trip
- `range(1, 13)` stops at 12
- Checks in the wrong order: put the strictest first